

$$3. b \geq a \geq c \quad \min(b, c) = c \quad \min(a, \min(b, c)) = \min(a, c) = c$$

$$\min(a, b) = a \quad \min(\min(a, b), c) = \min(a, c) = c$$

$$\therefore \min(a, \min(b, c)) = \min(\min(a, b), c)$$

$$4. b \geq c \geq a \quad \min(b, c) = c \quad \min(a, \min(b, c)) = \min(a, c) = a$$

$$\min(a, b) = a \quad \min(\min(a, b), c) = \min(a, c) = a$$

$$\therefore \min(a, \min(b, c)) = \min(\min(a, b), c)$$

$$5. c \geq a \geq b \quad \min(b, c) = b \quad \min(a, \min(b, c)) = \min(a, b) = b$$

$$\min(\min(a, b), c) = \min(b, c) = b$$

$$\therefore \min(a, \min(b, c)) = \min(\min(a, b), c)$$

$$6. c \geq b \geq a \quad \min(b, c) = b \quad \min(a, \min(b, c)) = \min(a, b) = a$$

$$\min(\min(a, b), c) = \min(a, c) = a$$

$$\therefore \min(a, \min(b, c)) = \min(\min(a, b), c)$$

$\therefore$ We have shown that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a, b and c are real numbers. $\square$

7. Prove using the notion of without loss of generality that $\min(x, y) = (x+y - |x-y|)/2$ and $\max(x, y) = (x+y + |x-y|)/2$ whenever x and y are real no.s

Solution: Let, x and y be two arbitrary real no.s. Since, they are arbitrary, we can combine the cases $x \geq y$ and $y \geq x$ into a single case.

Case 1: Let, wlog, $x \geq y$. $\therefore \min(x, y) = y = (2y)/2 = (x+y) - (x-y)/2$

$$= (x+y - |x-y|)/2 \quad (\because x-y \geq 0)$$

$$\max(x, y) = x = (2x)/2 = (x+y + (x-y))/2 = (x+y + |x-y|)/2 \quad (\because x-y \geq 0)$$